PROBLEM 2 — THE PAPER PLANE CHALLENGE

Aerodynamic Analysis and Experimental Performance Report ("Suzanne" Glider)

PART A — DESIGN

Wing Shape, Dimensions & Configuration Justification: Standard paper darts pitch down rapidly due to small wing area and high stall speed. The chosen Suzanne design utilizes a broad wingspan to achieve a high aspect ratio (wingspan squared divided by total wing surface area), which minimizes induced drag at low speeds. Maximizing surface area lowers the wing loading (total weight divided by planform area), directly reducing minimum flying speed and sink rate. Upward vertical winglets reduce tip vortices by blocking high-pressure air under the wing from spilling into the low-pressure region above. A positive dihedral angle (wings angled slightly upward in a V-shape) provides self-righting roll stability during flight.

PART B — AERODYNAMICS

1. Lift Generation Mechanisms: (A) *Bernoulli's Principle:* Due to wing camber and angle of attack, oncoming air moves faster across the upper wing surface than underneath (top velocity exceeds bottom velocity). This creates lower static pressure above the wing, generating an upward pressure differential. (B) *Newton's Third Law:* The inclined wing deflects an oncoming mass of air downward (downwash). By action-reaction, an equal and opposite upward force acts on the wing.

2. Force Analysis Across Flight Phases: *Launch Phase:* Manual throw provides initial kinetic energy (thrust exceeds drag). High launch speed creates rapid climb as kinetic energy converts to gravitational potential energy. *Glide Phase:* Unpowered flight (sustained thrust equals zero). Gravity pulls the aircraft down at a shallow glide angle. The forward component of gravity acts as the propulsive force against drag, while the perpendicular component is balanced by lift. The glide slope angle equals drag divided by lift.

3. Stability & Aerodynamic Parameters: *Wing Loading:* Lower wing loading reduces stall speed, keeping the plane airborne at slower forward airspeeds. *Center of Gravity (CG) vs. Center of Lift (CL):* Pitch stability requires CG to sit forward of CL. The resulting nose-down pitching moment is trimmed in level flight by upward micro-bends (up-elevators) on the trailing wing edges. *Dihedral Angle:* Sideslip during roll generates higher relative lift on the lower wing, restoring the plane to wings-level equilibrium.

4. Minimum Sink Rate vs. Maximum Glide Distance: Distance-optimized planes maximize the Lift-to-Drag ratio. Hang-time planes minimize vertical sink speed (rate of altitude loss). Minimum sink rate requires maximizing the power factor (lift coefficient raised to the power of 1.5 divided by drag coefficient) at slow airspeeds, rather than minimizing drag alone at higher speeds.

PART C — EXPERIMENT

1. Flight Trial Data:

Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Mean Flight Duration
7.00 s	7.00 s	8.00 s	8.00 s	9.00 s	**7.80 seconds**

2. Suggested Modifications & Controlled Testing (Part C, Item 3): (A) *Micro Up-Elevator Flaps:* Bending trailing edges upward by 1 mm increases downforce at the rear, increasing angle of attack toward the optimal power factor. Tested over 5 trials holding throw velocity constant. (B) *Winglet Height Adjustment:* Increasing winglet height from 1 cm to 1.5 cm further isolates pressure regions across the wingtip, reducing induced drag. Tested holding trim angle and launch conditions constant.